EGA: Adapting Frozen Encoders for Vector Search with Bounded Out-of-Distribution Degradation

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Abstract

Vector search systems built on frozen vision encoders face queries from unseen classes at deployment, yet existing adapter training collapses under this shift: high-capacity adapters with global contrastive losses silently reassign unseen-class samples to wrong seen-class clusters, dropping Label Precision by over 40 points below the frozen baseline. We propose Euclidean Geodesic Alignment (EGA), a residual adapter that couples three simple principles: zero initialization, local triplet loss, and hypersphere projection. These collectively induce a self-limiting dynamic: triplets that already satisfy a small margin stop producing gradients, so the adapter automatically stops updating where the local geometry is already correct. At convergence 96.5% of triplets are gradient-free, leaving unseen-class regions largely untouched while still enabling full-capacity refinement of seen classes. Across four diverse OOD benchmarks, EGA achieves the highest worst-case Label Precision (0.611): the only method to exceed 0.6 on every single benchmark. The design transfers directly to stronger backbones, and we provide an analytical justification linking gradient sparsity to bounded OOD perturbation.

1 Introduction

A vector similarity search system [16, 30] deployed in production rarely operates on a fixed label space. A retailer indexes product images today and adds a new category line next month; a content platform builds an index over existing tags and continually ingests media from emerging topics; a scientific image database is annotated for known phenotypes and queried for ones discovered later. In such scenarios, the index and the query stream together cover a label space strictly larger than what was available when any model was trained or adapted.

The deployment reality of unseen classes sits awkwardly with how adapters over frozen encoders [25] are currently trained. The dominant paradigm takes a labeled subset, optimizes the embedding geometry to separate those classes, and ships the resulting adapter as a drop-in replacement for the frozen encoder's output. The implicit assumption is that geometry tightened for seen classes will, at worst, leave unseen-class geometry unchanged. We show this assumption fails systematically: high-capacity adapters trained with global contrastive objectives reshape the entire embedding distribution, pulling unseen-class samples toward whichever seen-class clusters they happen to resemble in the pre-trained metric. The retrieval system inherits this distortion silently, since approximate nearest neighbor (ANN) indices remain well-behaved geometrically while the neighbors belong to the wrong semantic class.

The system designer usually cannot know in advance which out-of-distribution (OOD) will dominate the deployed query stream. The relevant criterion is therefore the *worst-case Label Precision* across plausible OODs, not optimality on any single one. Two design choices in current adapter training degrade this worst-case quantity. First, the loss function: global contrastive objectives such as

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ICon [1] and SRL [4] compute gradients over the full pairwise similarity matrix in every step, so seen-class supervision exerts pressure across the entire embedding distribution rather than only on labeled regions. Second, the adapter capacity: a sufficiently expressive adapter has the representational room to comply with that pressure, and zero-initialized residual designs do not by themselves prevent the cumulative perturbation from reaching unseen-class regions. Existing alternatives address one factor at a time. Low-rank adapters such as LoRA [10] cap the second factor, which avoids the worst OOD failures but limits in-distribution (ID) refinement, while keeping the global loss leaves the first factor untouched. We argue that adapters intended to operate under unseen-class queries should be designed against both factors in a coordinated manner.

The cost of failing on either factor is large, as we will demonstrate in Section 4. On the strictly unseen classes of Food-101, high-capacity global contrastive adapters collapse to 0.562 and 0.552 Label Precision, between 32 and 33 percentage points below the frozen CLIP baseline. On the cross-dataset CIFAR-10 OOD setting the same methods drop 32 to 34 percentage points below the baseline. LoRA variants avoid this collapse but pay for it in-distribution, with Label Precision bounded between 0.61 and 0.67 on CIFAR-100, well below what a high-capacity adapter can achieve when the loss does not over-extend into unseen-class regions.

This work presents Euclidean Geodesic Alignment (EGA), a residual adapter designed against both factors of loss function and adapter capacity, simultaneously. EGA pairs a high-capacity zero-initialized residual architecture with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings, with ℓ_2 projection onto the unit hypersphere. Despite the technical intricacies, the underlying idea is straightforward, as follows. On the one hand, the triplet loss induces a self-limiting training dynamic: as the adapter converges on seen classes, the fraction of margin-violating triplets decays rapidly. On the other hand, the capacity to refine seen-class neighborhoods is preserved, but the loss stops applying pressure once seen-class neighborhoods are locally separated. As a result, the global reshaping that destroys unseen-class geometry under ICon and SRL never accumulates.

We evaluate EGA on CIFAR-100 and ImageNet-1K (in-distribution) and four OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101, ImageNet-1K held-out classes), comparing against the frozen CLIP baseline [25], global contrastive adapters (ICon [1], SRL [4]), and LoRA [10] variants (InfoNCE [32] and Triplet, both at $r = 128$). EGA achieves the highest worst-case OOD Label Precision (0.611) across the four OOD benchmarks, 4.2 percentage points above the next-best method (LoRA+Triplet at 0.569). EGA is the only method whose Label Precision exceeds 0.6 on every OOD benchmark. ICon and SRL collapse to 0.470 and 0.375 in their worst case (Aircraft); LoRA+InfoNCE to 0.538 (Aircraft). The benefits transfer to stronger backbones, with EGA improving Label Precision@1 by 4.75 and 8.55 percentage points on DINOv2-large [23] and SigLIP [39]. EGA does not dominate every method on every individual OOD benchmark: on CIFAR-10, Food-101, and ImageNet-1K held-out classes, capacity-matched LoRA achieves higher Label Precision than EGA on the easier OOD shifts, but trades this for catastrophically lower worst-case quality on harder shifts.

In summary, this work makes the following contributions:

- We frame adapter tuning over frozen vision-language encoders from a perspective of vector search applications, in which the index and query stream together cover a label space larger than the labeled subset available for adapter training, and we identify worst-case OOD Label Precision as the relevant criterion under deployment uncertainty.

- We propose Euclidean Geodesic Alignment (EGA), a residual adapter design that couples zero-initialized residual architecture, local triplet supervision, and hypersphere projection. This combination induces a self-limiting training dynamic: as seen-class neighborhoods become locally separated, the fraction of active triplets decays rapidly, empirically reaching 96.5% zero-gradient triplets at steady state. We also provide a theoretical analysis of this dynamic, showing that the unseen-class perturbation is bounded by the product of the active triplet ratio and the adapter's Lipschitz constant.

- Across four OOD benchmarks and three backbones (CLIP ViT-B/32, DINOv2-large, SigLIP), EGA achieves the highest worst-case OOD Label Precision (0.611), exceeding all five baselines by between 4.2 and 23.6 percentage points. EGA is the only method whose Label Precision exceeds 0.6 on every OOD benchmark we evaluate.

2 Background and Related Work

Foundation models such as CLIP [25] produce embedding spaces that encode both seen-class semantics and broader priors over unseen visual concepts. Related work has documented degradation phenomena in adapted representations [20, 37], and recent work documents analogous degradation in contrastive representations, including anisotropic collapse [43, 6], modality gap fracturing [26, 5], and loss of compositional structure [24, 28, 36]. Recent strategies counter these issues with geometric constraints such as orthogonality regularization [27]. *EGA exploits the gradient sparsity of small-margin triplet losses to keep adapter updates concentrated on seen-class neighborhoods, empirically leaving unseen-class regions of the pretrained geometry largely intact.*

Recent representation learning frequently relies on global contrastive objectives to optimize the latent space [31, 19]. A common paradigm jointly enforces alignment for positive pairs and uniformity across the feature distribution [34], applied in graph encoders [33] and multimodal recommendation [46]. While uniformity improves robustness to noise [35], it treats all latent regions equally. Other structural regularizers include unifying frameworks for multi-view alignment [1], geometric constraints for imbalanced regression [4], decoupled objectives [18], information-theoretic density control [42], and region-based distillation [15]. *EGA replaces dense global optimization with sparse local triplet updates, which empirically reduces unseen-class distortion.*

Parameter-efficient adaptation is an approach for transferring large foundation models without full fine-tuning [9, 10], which dominates vision-language research [21]. Lightweight methods, e.g., Visual Prompt Tuning [14] and Tip Adapter [41], specialize the model without modifying the frozen backbone and rival full fine-tuning on many tasks [29], while zero-initialization strategies further constrain the magnitude of pretrained-manifold perturbation [40]. *EGA admits a zero-initialized residual adapter, localized triplet objective, and parameter efficiency with gradient-sparse updates.*

Vector search is the foundational engine for scalable high-dimensional retrieval. The systems community has developed indexing algorithms for billion-scale datasets, including FAISS [16], HNSW [22], DiskANN [30], and SPANN [2]. Recent work refines graph topologies [38, 3], optimizes SSD-based search [7, 8], and accelerates distance computations [13], yet popular ANN implementations still suffer worst-case limitations [12]. *EGA recalibrates the metric geometry before index construction, rendering search backends achieve higher semantic recall without algorithmic changes.*

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \to \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$. Our goal is to learn an adapter $f_\theta : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to unseen-class labels during training.

We evaluate retrieval quality along two complementary dimensions. *Label Precision@K* (LP@K) measures the semantic quality of retrieved results: whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\left[y_{\hat{n}_k(q)} = y_q\right], \tag{1}$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k-th nearest neighbor of q in the transformed embedding space. *ANNS Recall@K* (AR@K) measures the geometric fidelity of the ANN index, i.e, the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget nprobe:

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{\left|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)\right|}{K}, \tag{2}$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top-K neighbor sets, respectively. The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@K by making the geometry more amenable to IVF-based indexing (geometric improvement).

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is constrained by three design principles, each of which contributes to keeping the adapter's updates anchored to the pretrained geometry.

(1) Residual connection with zero initialization. Each EGA block applies a residual update as:

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \tag{3}$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding.

(2) Feature gating. Inside each residual block, the transformed signal passes through a module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2\,\text{GELU}(W_1\mathbf{h}))\,, \tag{4}$$

where $W_1 \in \mathbb{R}^{(d/4)\times d}$, $W_2 \in \mathbb{R}^{d\times(d/4)}$, σ is the Sigmoid function, and $\odot$ denotes elementwise multiplication. This is analogous to Squeeze-and-Excitation networks [11] to learn a per-dimension weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio $1/4$ is chosen to prevent the gating from overfitting on seen-class data, to be validated in Table 4.

The output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \tag{5}$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original embeddings: Euclidean distance and cosine similarity remain interchangeable on the hypersphere.

(3) Full forward pass. Let Block_i denote the i-th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm. The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z}))))\,. \tag{6}$$

The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension 2048, which is negligible compared to the frozen CLIP backbone ($\sim$88M parameters).

3.3 Local Triplet Training

Given a mini-batch $\mathcal{B}$ sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a, and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|}\sum_{(a,p,n)\in\mathcal{B}} \max(0,\ d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m)\,, \tag{7}$$

where $d(\cdot,\cdot)$ denotes distance, $m > 0$ is the margin, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding.

A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \tag{8}$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from

$\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter's updates remain effectively local (i.e., the cumulative perturbation induced by seen-class training gradients remains small in unseen-class regions of the embedding space), the manifold structure for unseen classes is largely preserved. The gradient sparsity mechanism enforces this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

The triplet loss in Eq. (7) consists of hinge terms whose subgradient with respect to θ is exactly zero whenever the margin constraint is satisfied:

$$\nabla_\theta \ell_{apn} = \mathbf{1}[d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0] \cdot \nabla_\theta \delta_{apn}, \tag{9}$$

where $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. Define the *active triplet ratio* at training step t as

$$\rho(t) = \Pr_{(a,p,n)\sim\mathcal{B}}\Big[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0\Big]. \tag{10}$$

Under the standard assumption that training converges to a stationary point of $\mathcal{L}$, $\rho(t)$ converges to a steady-state value $\rho^* \in [0, 1]$ determined by the margin m and the within-class distance distribution of the converged embeddings. We measure $\rho^* \approx 3.5\%$ at convergence on FGVC-Aircraft (Section 4.3), meaning that 96.5% of training triplets contribute zero gradient at steady state.

Strictly speaking, sparse seen-class gradients do not formally imply that f_θ leaves unseen-class regions unchanged: the adapter is a parametric MLP, not a locally supported transformation, so any update to θ propagates globally. The OOD measurements in Section 4.2 will demonstrate the evidence: EGA preserves or improves Label Precision on three unseen-class benchmarks, while ICon and SRL lack the gradient sparsity and fall below the frozen baseline. A Lipschitz-based bound on the unseen-class perturbation is discussed in Appendix A.

4 Evaluation

Experiments were conducted on Chameleon Cloud [17] with 256 AMD EPYC-7763 CPU cores, 512 GiB RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Datasets include CIFAR-100 and ImageNet-1K for in-distribution scenarios, CIFAR-10 for cross-dataset OOD settings, and FGVC-Aircraft, Food-101, and ImageNet-1K held-out classes for unseen-class OOD evaluations. All datasets can be downloaded from PyTorch or Kaggle. The backbone models include CLIP ViT-B/32 [25], DINOv2-large [23], and SigLIP [39].

We compare EGA against four adapter methods: ICon [1], SRL [4], LoRA + InfoNCE [10, 32], and LoRA + Triplet [10]. All adapters are trained on the same frozen CLIP features, under identical data splits and evaluation protocols. ICon and SRL are implemented as capacity-matched residual adapters (approximately 4.2M parameters each), differing from EGA only in their training objective; this isolates the effect of the loss function and ensures that performance differences are not driven by representational budget. The LoRA variants use rank r=128, also matching the 4.2M parameter budget, with zero-initialized residual connections. Full training details are in Appendix B.

We omit linear probing, Tip-Adapter [41], and prompt tuning methods (CoOp/CoCoOp [45, 44]) because they are not compatible with our cross-dataset OOD setting. Extended experimental results, including those for some of the omitted baselines, can be found in Appendix C.

4.1 In-Distribution Validation

Figure 1 reports Label Precision@K across four neighborhood sizes and three search budgets on CIFAR-100. EGA raises Label Precision@1 from 0.549 (raw CLIP) to 0.705 at nprobe $= 1$, and the gap persists across all K and nprobe values, including nprobe $= 10$, indicating that the adapter tightens within-class neighborhoods rather than benefiting from broader index coverage.

The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top-K neighbors against random background points. In the raw CLIP space, the two distributions overlap substantially, which is the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K, and this separation drives the recall-efficiency improvement in Figure 3: raw CLIP achieves 0.667 Recall@1 at nprobe $= 1$, whereas EGA reaches 0.825 and saturates above 0.99 by nprobe $= 5$.

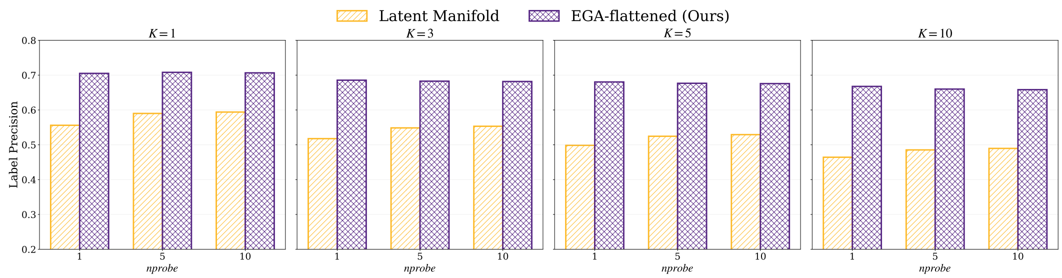

Figure 1: Label Precision@K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and nprobe $\in \{1, 5, 10\}$.

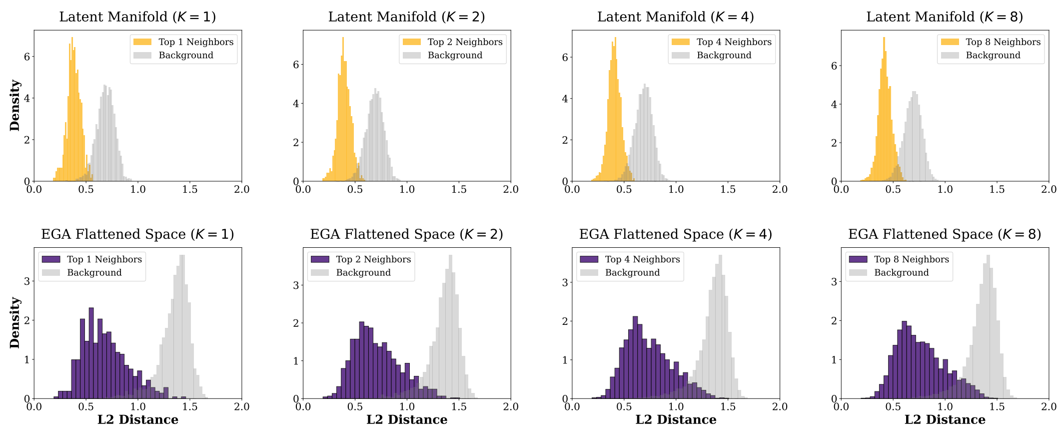

Figure 2: Distance distributions for Top-K neighbors vs. random background points on CIFAR-100.

Table 1 reports all six methods' in-distribution performance on both CIFAR-100 and ImageNet-1K. On CIFAR-100, ICon and SRL reach near-perfect Label Precision (0.999 and 0.992) by maximally tightening seen-class clusters under their global contrastive objectives. EGA reaches 0.705, an improvement of +16 percentage points over the frozen baseline; the LoRA variants land between EGA and the baseline. On ImageNet-1K (1000 classes), the scale of all methods drops because the test set has approximately seven samples per class for retrieval (compared to ∼150 on CIFAR-100), but the relative ordering changes: LoRA+InfoNCE leads at 0.426, with SRL (0.421), EGA (0.407), ICon (0.386), LoRA+Triplet (0.359), and CLIP (0.321) trailing. The within-2-percentage-point clustering of the four trained adapters (SRL, ICon, EGA, LoRA+Triplet) at this density indicates that ID performance becomes test-set-density-bounded rather than method-bounded as the label space grows.

Table 1: In-distribution retrieval performance at $K = 1$, nprobe $= 1$ on CIFAR-100 and ImageNet-1K. Standard errors are below 0.01 and omitted.

Method	CIFAR-100		ImageNet-1K	
	LP@1	AR@1	LP@1	AR@1
CLIP (frozen)	0.549	0.667	0.321	0.803
ICon	**0.999**	**0.992**	**0.386**	**0.817**
SRL	**0.992**	**0.991**	**0.421**	0.687
LoRA+InfoNCE	**0.668**	**0.879**	**0.426**	**0.822**
LoRA+Triplet	**0.612**	**0.908**	**0.359**	**0.870**
EGA	**0.705**	**0.825**	**0.407**	**0.817**

4.2 Out-of-Distribution Generalization

We now examine performance under strict OOD settings where evaluation categories are disjoint from training categories. Table 2 reports headline numbers at $K = 1$, nprobe $= 1$ across four OOD benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), Food-101 (80 seen / 21 unseen classes), and ImageNet-1K held-out classes (800 seen / 200 unseen classes). For a fair comparison, all methods train identical adapters on top of frozen CLIP features and differ only in their loss functions. Robustness across search budgets is reported in Appendix C.

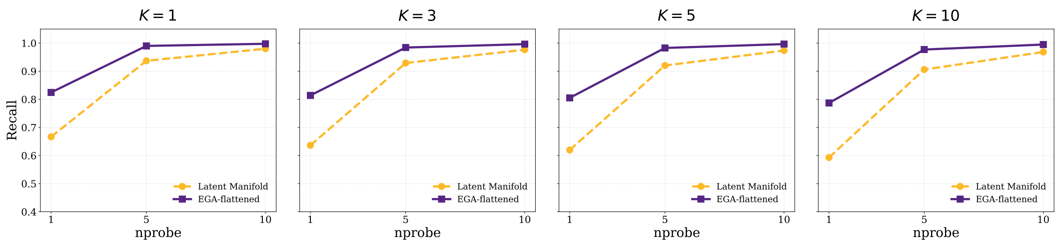

Figure 3: ANNS Recall@K versus nprobe on CIFAR-100.

Table 2: OOD performance at $K = 1$, nprobe $= 1$.

Dataset	Metric	CLIP	ICon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	0.797±.003	0.819±.001	**0.869**±.012	0.861±.011	0.833±.005
	LP@1	0.880	0.560±.010	0.536±.008	**0.885**±.006	0.875±.006	0.810±.002
Aircraft	AR@1	0.773	**0.853**±.016	**0.879**±.023	**0.891**±.015	**0.893**±.008	**0.905**±.015
	LP@1	0.512	0.470±.034	0.375±.032	**0.538**±.020	**0.569**±.011	**0.611**±.020
Food-101	AR@1	0.842	0.821±.001	0.824±.015	**0.906**±.003	**0.893**±.008	**0.879**±.011
	LP@1	0.881	0.562±.011	0.552±.010	0.883±.004	0.833±.007	0.791±.002
ImageNet-1K	AR@1	0.829	0.849±.012	0.747±.015	**0.878**±.008	**0.883**±.009	**0.868**±.011
	LP@1	0.684	0.620±.014	0.665±.012	**0.753**±.009	**0.714**±.010	**0.711**±.008
Worst-case	**LP@1**	0.512	0.470	0.375	0.538	0.569	**0.611**

Worst-case OOD Label Precision. The bottom row of Table 2 reports the worst-case Label Precision across the four OOD benchmarks. EGA achieves 0.611, exceeding LoRA+Triplet (0.569) by 4.2 percentage points, LoRA+InfoNCE (0.538) by 7.3 percentage points (pp), the frozen CLIP baseline (0.512) by 9.9 pp, ICon (0.470) by 14.1 pp, and SRL (0.375) by 23.6 pp. EGA is the only method whose Label Precision exceeds 0.6 on every OOD benchmark we evaluate.

The four OOD benchmarks span qualitatively different shifts: CIFAR-10 is a cross-dataset shift; Aircraft and Food-101 are fine-grained intra-domain shifts where seen and unseen classes share substantial visual structure; ImageNet held-out classes is a generic large-scale shift where the pretrained encoder already provides reasonable transfer. The worst-case reflects deployment uncertainty: the system designer commits to one adapter without knowing in advance which shift will dominate the query stream. Methods that excel on some shifts but collapse on others (ICon collapses on Aircraft, SRL collapses on Aircraft, LoRA variants on Aircraft) cannot guarantee a quality floor.

Per-benchmark behavior. On ANNS Recall, all trained adapters match or exceed the frozen baseline across Aircraft, Food-101, and ImageNet, confirming that adapter training successfully reorganizes the geometry for ANN indexing. On Label Precision, however, the global contrastive methods fall below the frozen baseline on every fine-grained OOD dataset, with degradation exceeding 30 percentage points on Food-101 and CIFAR-10: the retrieved neighbors are geometrically closer but semantically wrong. EGA improves Label Precision over the frozen baseline on Aircraft and ImageNet, and its degradations on CIFAR-10 and Food-101 are contained to single-digit percentage points. The collapse of global contrastive methods is most severe exactly where unseen classes are visually similar to seen classes, consistent with our hypothesis that dense global gradients pull unseen samples into incorrect seen-class clusters. On ImageNet, where held-out classes are visually distinct from training classes, no method collapses, and all approaches, including the frozen baseline, perform comparably. EGA's advantage materializes precisely under difficult distribution shifts, which is the scenario where worst-case guarantees are most valuable.

EGA does not dominate every method on every individual benchmark. Methods that restrict adapter capacity rather than the loss function achieve higher Label Precision on CIFAR-10, Food-101, and ImageNet, but suffer catastrophic degradation on the fine-grained Aircraft split, which is what determines their worst-case performance. EGA's design explicitly prioritizes worst-case robustness;

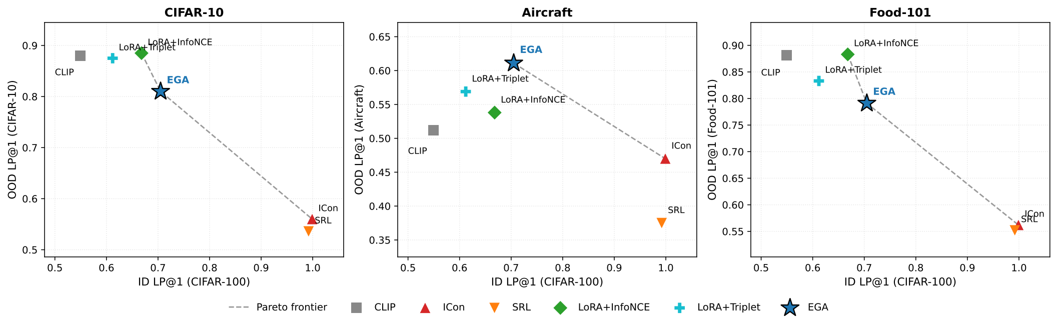

Figure 4: ID/OOD operating points of adapter training in the (ID, OOD) Label Precision plane.

practitioners who can confidently bound the deployed distribution shift may reasonably prefer capacity-restricted alternatives. The trade-off between these two approaches is visualized in Figure 4.

4.3 Two Routes to OOD Stability: Gradient Sparsity and Capacity Restriction

Adapters that avoid OOD collapse must neutralize one of these factors, and the route determines its position in the Label Precision plane.

Route 1: Gradient sparsity (EGA). A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated:

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

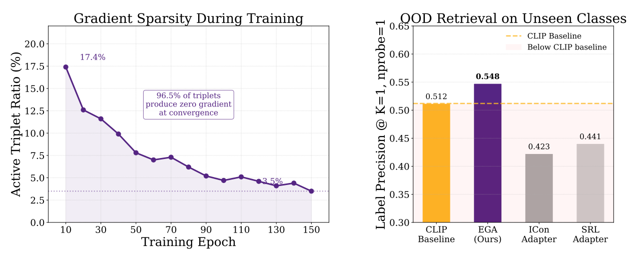

Figure 5: Gradient sparsity as OOD protection.

As the adapter converges on seen classes, the fraction of *active* triplets decays. Figure 5 (left) tracks this ratio from 17.4% at epoch 10 to 3.5% at convergence. At steady state, 96.5% of triplets contribute zero gradient: the loss stops applying pressure once seen-class neighborhoods are locally separated, even though the adapter retains the capacity to refine them further. Sparsity neutralizes the loss-side factor while leaving capacity intact, which permits EGA's high in-distribution Label Precision in Table 1.

Route 2: Capacity restriction (LoRA). The LoRA variants take the other route. They retain a global contrastive loss (InfoNCE) or a triplet loss, but constrain the adapter to a low-rank residual update. At rank $r = 128$, LoRA's expressive power is limited to rank-128 perturbations of the identity, so the dense gradient field has substantially less room to reshape unseen-class geometry. This explains why LoRA+InfoNCE, despite using the same loss family as ICon, avoids the same OOD collapse: the loss applies pressure everywhere, but the adapter cannot fully comply. The price is paid in-distribution. At capacity-matched scale ($\sim$4.2M parameters), LoRA+InfoNCE reaches 0.668 Label Precision on CIFAR-100, 3.7 percentage points below EGA, because the same low-rank constraint that bounds OOD damage also bounds in-distribution refinement.

The two routes occupy different points in Figure 4. EGA, which neutralizes the loss-side factor while keeping capacity, achieves the highest worst-case OOD Label Precision because it does not collapse on any single OOD shift. Capacity-matched LoRA, which neutralizes the capacity-side factor while keeping a global loss, achieves higher OOD Label Precision on three of four benchmarks (CIFAR-10, Food-101, ImageNet) but collapses on the fine-grained Aircraft benchmark, capping its worst-case at 0.538 to 0.569. Whether either route subsumes the other under stronger conditions, for example by combining gradient sparsity with low-rank constraints, is a question we leave open.

Table 3: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.705
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 4: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

4.4 Ablation and Sensitivity Analysis

Table 3 isolates the contribution of each architectural component. The residual connection is the core element: removing it collapses accuracy from 0.705 to 0.0065, indicating that the adapter cannot reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. ℓ_2-normalization keeps the transformed embeddings on the unit hypersphere, and removing it lowers accuracy to 0.6590. Zero-initialization lets the adapter begin as an identity mapping, and removing it lowers accuracy to 0.6840.

Table 4 evaluates stability across hyperparameter variations. EGA maintains consistent quality for margins $m \in \{0.1, 0.2, 0.3\}$, but increasing m to 0.5 degrades Recall@1 to 0.7400: a larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, reducing gradient sparsity. Performance is also stable across hidden dimensions from 512 to 4096, indicating that EGA's behavior is determined by its structural design rather than by specific parameter choices.

4.5 Generalization to More Backbones

To verify that EGA's benefits are not specific to CLIP ViT-B/32 [25], we further evaluate it on two more recent vision encoders: DINOv2-large [23] and SigLIP [39].

Table 5 shows consistent improvements across all three backbones. The largest gains appear on the relatively weaker CLIP ViT-B/32 ($+13.95$ pp in Label Precision@1 and $+13.20$ pp in Recall@1), but the stronger DINOv2-large and SigLIP backbones also see stable improvements ($+4.75$ pp and $+8.55$ pp in Label Precision@1 respectively). EGA's design therefore transfers to modern high-capacity vision encoders rather than relying on idiosyncrasies of the CLIP representation.

Table 5: Retrieval performance on more backbones.

Backbone	Label Precision@1		ANNS Recall@1	
	Raw	+EGA	Raw	+EGA
CLIP ViT-B/32	0.5560	**0.6955**	0.6665	**0.7985**
DINOv2-large	0.8530	**0.9005**	0.8785	**0.9310**
SigLIP SO400M	0.7740	**0.8595**	0.8015	**0.9155**

5 Conclusion

We studied adapter training for vector search under open-category queries, where the proper metric is worst-case OOD Label Precision rather than in-distribution optimality. Global contrastive adapters silently distort unseen-class geometry; the proposed method, namely EGA, avoids this through a self-limiting triplet loss that vanishes when local class structure is satisfied, automatically preserving unseen regions. Across diverse shifts, EGA achieves the highest worst-case retrieval quality, is the only method above 0.6 on every benchmark, and transfers directly to multiple backbones. We link gradient sparsity to bounded geometric drift and propose the active triplet ratio as a diagnostic for adapters that must generalize to unseen classes for vector search applications.

Limitations First, our theoretical bound depends on the empirically observed steady-state active triplet ratio; a closed-form guarantee without this empirical input remains open. Second, EGA prioritizes worst-case robustness, so it does not dominate every benchmark: methods that restrict capacity can excel on easier shifts, and practitioners who know their deployment shift may prefer them. Third, while our loss-function comparisons control for adapter capacity, the LoRA variants additionally alter the adapter structure (low-rank), so the effects of loss type and architectural constraint are not fully isolated; a factorial study under strictly matched conditions would further clarify their individual contributions.

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A Theoretical Analysis of EGA's OOD Stability

We provide a theoretical analysis to explain why the proposed EGA adapter outperforms other popular adapters in worst-case out-of-distribution (OOD) scenarios. The analysis is structured in four steps, starting from the geometric interpretation of the adapter architecture to deriving a Lipschitz-based bound on the unseen-class perturbation.

1. Manifold Projection and Identity Initialization

Let the pre-trained feature space be defined as a unit hypersphere manifold $\mathcal{M} = \mathbb{S}^{d-1} \subset \mathbb{R}^d$. For any input feature $x \in \mathcal{M}$, the residual adapter learns a continuous perturbation vector field $h_\theta : \mathcal{M} \to \mathbb{R}^d$. The forward pass of our architecture, which combines a residual connection with L_2 normalization, is geometrically equivalent to a retraction mapping $\Pi_{\mathcal{M}}$ that projects the perturbed point back onto the manifold:

$$f_\theta(x) = \Pi_{\mathcal{M}}(x + h_\theta(x)) = \frac{x + h_\theta(x)}{\|x + h_\theta(x)\|_2}. \tag{12}$$

Under our zero initialization strategy, the initial perturbation field is strictly zero everywhere, meaning $h_{\theta_0}(x) \equiv 0$. Consequently, the initial adapter state represents a strict identity mapping on the manifold, $f_{\theta_0}(x) = x$. This forms the theoretical origin for computing cumulative structural distortions.

2. Path Integral of Unseen Class Distortion

We model the training process as continuous gradient descent where parameters evolve according to $\frac{d\theta_t}{dt} = -\eta \nabla_\theta \mathcal{L}(\theta_t)$, with learning rate η. For any given unseen class sample $x_u \in \mathcal{X}_{unseen}$, its total geometric drift accumulated over the training period T can be formulated using the fundamental theorem of calculus as a path integral along the parameter trajectory:

$$\Delta(x_u) = \|f_{\theta_T}(x_u) - f_{\theta_0}(x_u)\|_2 = \left\| \int_0^T J_{\theta_t}(x_u) \frac{d\theta_t}{dt} dt \right\|_2, \tag{13}$$

where $J_{\theta_t}(x_u) = \nabla_\theta f_{\theta_t}(x_u)$ denotes the Jacobian matrix of the network evaluated at x_u. Applying the triangle inequality for integrals, we establish the fundamental upper bound for geometric distortion:

$$\Delta(x_u) \leq \int_0^T \left\| J_{\theta_t}(x_u) \frac{d\theta_t}{dt} \right\|_2 dt. \tag{14}$$

3. Sparsity Modulated Deformation Bound

In our formulation, the parameter derivative is entirely driven by the active triplet ratio ρ_t. Let N be the batch size. The loss gradient is the average over active triplets. Assuming the gradient norm per individual sample is bounded by G, the batch gradient norm satisfies:

$$\|\nabla_\theta \mathcal{L}(\theta_t)\|_2 = \left\| \frac{1}{N} \sum_{i \in active} \nabla_\theta \ell_i \right\|_2 \leq \frac{1}{N} \sum_{i \in active} \|\nabla_\theta \ell_i\|_2 \leq \frac{\rho_t N}{N} G = \rho_t G. \tag{15}$$

This directly bounds the parameter update rate:

$$\left\| \frac{d\theta_t}{dt} \right\|_2 = \eta \|\nabla_\theta \mathcal{L}(\theta_t)\|_2 \leq \eta \rho_t G. \tag{16}$$

To process the integrand from the previous step, we apply the sub-multiplicativity of matrix norms. Assuming the network satisfies L Lipschitz continuity such that the spectral norm of the Jacobian is bounded by $\|J_{\theta_t}(x_u)\|_2 \leq L$, we obtain:

$$\left\| J_{\theta_t}(x_u) \frac{d\theta_t}{dt} \right\|_2 \leq \|J_{\theta_t}(x_u)\|_2 \left\| \frac{d\theta_t}{dt} \right\|_2 \leq L(\eta \rho_t G). \tag{17}$$

Substituting this back into the path integral yields the global perturbation upper bound for any unseen sample:

$$\Delta_{EGA}(x_u) \leq \eta L G \int_0^T \rho_t dt. \tag{18}$$

For global contrastive methods, the equivalent active ratio remains at a constant level near 1, resulting in a perturbation bound that grows linearly with time ηLGT. In contrast, our gradient sparsity mechanism forces ρ_t to decay rapidly to a marginal steady state value. The bounded nature of this integral mathematically locks the maximum possible geometric shift for unseen categories.

4. Orthogonal Collapse of the Active Jacobian Subspace

To explain why the residual active gradients do not coincidentally distort visually similar unseen classes, we analyze the dimensionality of the update space. The parameter update direction $\frac{d\theta_t}{dt}$ resides entirely within a tangent subspace $\mathcal{S}_{active}^{(t)}$ spanned by the gradients of the active triplets. Let $P_{\mathcal{S}}$ denote the orthogonal projection operator onto this active subspace. We can rewrite the update vector as $P_{\mathcal{S}_{active}^{(t)}} \frac{d\theta_t}{dt}$. The true instantaneous perturbation is therefore modulated by the projection of the unseen Jacobian onto this subspace:

$$\left\| J_{\theta_t}(x_u) \frac{d\theta_t}{dt} \right\|_2 = \left\| J_{\theta_t}(x_u) P_{\mathcal{S}_{active}^{(t)}} \frac{d\theta_t}{dt} \right\|_2 \leq \left\| J_{\theta_t}(x_u) P_{\mathcal{S}_{active}^{(t)}} \right\|_2 \left\| \frac{d\theta_t}{dt} \right\|_2. \tag{19}$$

As the active triplet ratio ρ_t drops exponentially, the dimensionality of $\mathcal{S}_{active}^{(t)}$ collapses. Consequently, the spectral norm of the projected Jacobian $\left\| J_{\theta_t}(x_u) P_{\mathcal{S}_{active}^{(t)}} \right\|_2$ probabilistically approaches zero. Under the empirically observed decay of ρ_t, the dimensionality of $\mathcal{S}_{active}^{(t)}$ collapses, and the spectral norm of the projected Jacobian probabilistically approaches zero. This provides geometric intuition for why the sparse updates are unlikely to systematically corrupt the structure of unseen classes.

B Experimental Setup Details

Backbones models. All experiments in the main paper use frozen CLIP ViT-B/32 [25] image embeddings as the base representation. To verify that EGA's benefits are not specific to this backbone, we further evaluate EGA on two stronger vision encoders: DINOv2-large [23] and SigLIP [39]. DINOv2-large is a self-supervised vision transformer trained on 142M images, while SigLIP is a supervised vision transformer trained on 400M image-text pairs. Both models produce higher-quality embeddings than CLIP ViT-B/32, as reflected in their stronger raw retrieval performance, and EGA continues to provide meaningful improvements on top of these stronger backbones, as shown in Table 5 in Section 4.4.

Adapter models. We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **ICon** [1]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution.
- **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.
- **LoRA + InfoNCE** [10, 32]: a capacity-matched low-rank adapter (rank $r = 128$, $\sim$4.2M parameters) with zero-initialized residual structure, trained with supervised InfoNCE.
- **LoRA + Triplet** [10]: identical capacity-matched low-rank architecture as above (rank $r = 128$), but trained with the same small-margin ($m = 0.2$) triplet loss as EGA.

Datasets. We evaluate on five datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as a primary in-distribution benchmark with a 75/25 split between database and query sets. **ImageNet-1K** (1000 classes, 35K embedded samples) serves as a larger-scale in-distribution benchmark and additionally provides an unseen-class OOD setting via an 80/20 class split (800 seen / 200 unseen). **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split.

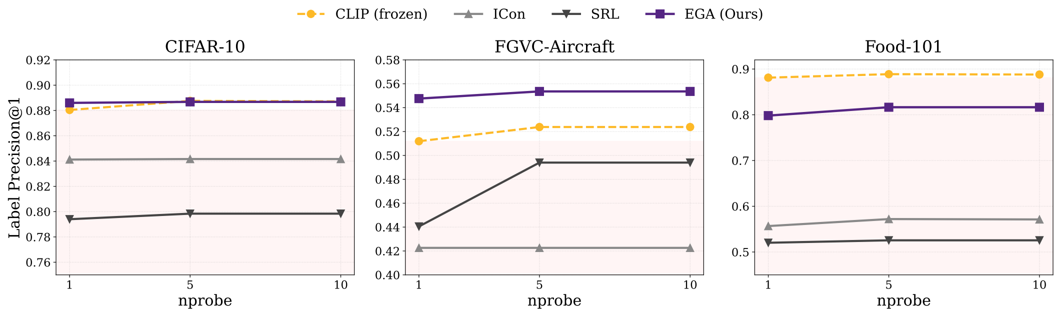

Figure 6: Label Precision@1 across three OOD benchmarks and three search budgets (nprobe $\in \{1, 5, 10\}$).

Indexing and metrics. For all retrieval evaluations we use FAISS [16] `IndexIVFFlat` with nlist=10 centroids and report results at nprobe $\in \{1, 5, 10\}$. We measure both Label Precision (LP@K, Eq. 1) and ANNS Recall (AR@K, Eq. 2) at $K \in \{1, 3, 5, 10\}$.

Training. All adapters are trained with AdamW (weight decay 10^{-4}), cosine-annealed learning rate, and batch size 256 (CIFAR-100) or 512 (ImageNet-1K). EGAMLP-based methods (EGA, ICon, SRL) use learning rate 10^{-4}; LoRA variants use 10^{-3} to compensate for their smaller per-parameter scale. Triplet-based methods sample one positive and one negative per anchor uniformly within each mini-batch; global contrastive methods (ICon, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix. All experiments use three fixed random seed (42, 123, 456) for reproducibility.

Compute platform. All experiments were conducted on the Chameleon Cloud [17] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.

C Additional Experimental Results

Robustness across search budgets. Figure 6 traces Label Precision@1 across three OOD benchmarks and three search budgets (nprobe $\in \{1, 5, 10\}$). The collapse pattern of ICon and SRL persists across all 18 measurement points (3 datasets $\times$ 3 budgets $\times$ 2 methods), ruling out the possibility that the degradation is an artifact of any particular benchmark or indexing configuration. EGA's relative ordering against other methods is also stable across budgets.

Linear Probing Baseline Analysis To address potential concerns about additional baselines, we also evaluated Linear Probing as an example. As shown in Table 6 and Table 7, Linear Probing achieves reasonable in-distribution performance (0.5112 LP@1 on CIFAR-100) but completely collapses on out-of-distribution benchmarks (0.0065–0.0133 LP@1). This confirms that Linear Probing is not a meaningful comparison in our cross-dataset OOD setting due to class semantic mismatch.

Table 6: Linear Probing in-distribution performance on CIFAR-100 (75/25 split, mean $\pm$ std over 3 seeds).

Method	LP@1	AR@1
Linear Probing	0.5112 $\pm$ 0.0020	0.7533 $\pm$ 0.0025

Table 7: Linear Probing OOD performance (mean $\pm$ std over 3 seeds).

Dataset	LP@1	AR@1
Aircraft	0.0133 ± 0.0038	0.4401 ± 0.0254
Food-101	0.0077 ± 0.0019	0.5162 ± 0.0228
CIFAR-10	0.0065 ± 0.0005	0.6454 ± 0.0096